

Shaper

- ① Shaper always used for shaping smaller jobs or machining smaller jobs.
- ② It is light duty machine.
- ③ It can only use one cutting tool at a time.
- ④ The working mechanism is quick return mechanism.
- ⑤ It is less rigid and less robust (tough, hard, having high strength).

Planner

- ① When the job size is as large as 6m wide and twice as long can be easily machined on planner.
- ② It is a heavy duty machine.
- ③ It can use several tools simultaneously.
- ④ They are working on gear arrangement and hydraulic arrangements.
- ⑤ As compared to shaper, it is more rigid and gives more accuracy on the machine surface.

Slotter

It is a machine which is considered as reciprocating type of machine and generally navigated as vertical shaper. It is always used for specific operations.

- ① Used for shaping internal or external blind holes.
- ② used for machining dies and punches.
- ③ Used for machining vertical surfaces.
- ④ Also used to machine angular or inclined surface.
- ⑤ It also helps to machine circular or semi-circular concave and convex surfaces.

In this, the tool & job is in stationary position and the tool moves on the ram in up and down direction. It is always used to create slots when the geometry of the object is very complicated. It is also considered as light-duty machine.

ch-4 → Center lathe

ch-5 → Special purpose lathe.

ch-6 → Reciprocating machine tools

ch-7 → Milling machine.

ch-11 → Unconventional machining process

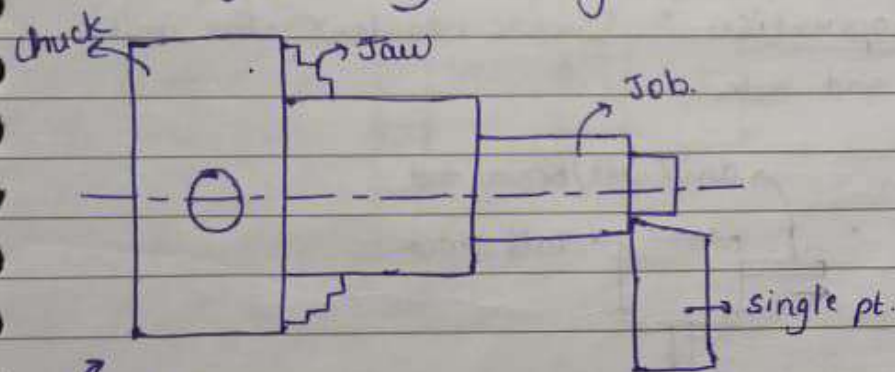
ch-14 → Jicks and fixture.

ch-16 → Numerical control of machine tools

Lathe operations

The working principle of lathe machine is job rotates and the cutting tool is stationary.

The various operations that can be performed on a lathe machines are turning, facing, knuddling, parting, drilling, ~~boreing~~ boring.



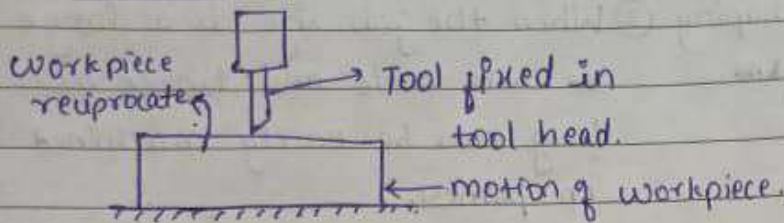
cutting operation line diag.

(i) Turning operation :-

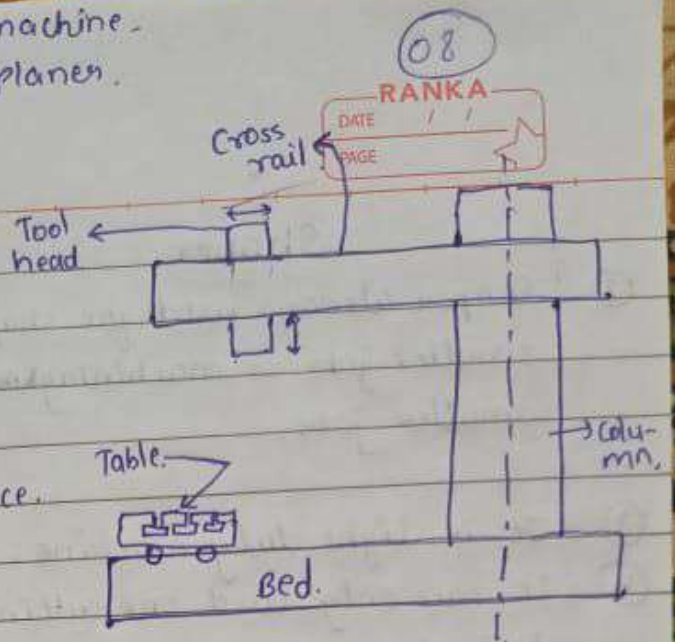
In this process, the workpiece held in the spindle is rotated while the tool is fed, pass the workpiece in a direction parallel to the axis of rotation. Generally, this process is used to remove the material, to create various cylindrical shapes.

Cylindrical object → lathe machine.
 Square → shaper, planer.

Planer :-



Working principle of a planer



Open side planer.

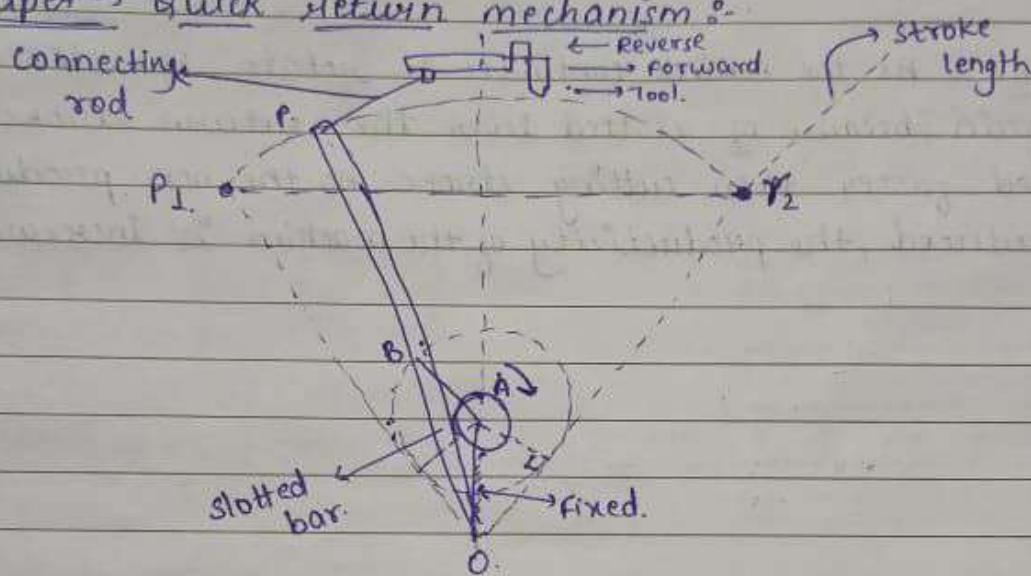
↳ When the diameter and length of an object is very large and it is not possible to machine ~~out~~ out on shaper, we move the job on the machine named planer. It is also used to produce horizontal, vertical or inclined flat surface with the help of single point cutting tool. It is always used for machining large and heavy workpieces. The working principle of planer is your cutting tool is stationary and the workpiece which is supported on the table reciprocates, it means tool is fixed and the work piece is in motion.

Que. Differentiate b/w shaper and planer.

S.No.	Shaper	Planer.
①	The workpiece is stationary & the cutting tool on the ram move in backward & forward direction across the workpiece.	The tool is stationary & the workpiece travels back and forth under the tool.

Chamfering : It is a process with the help of which the sharp corners from the object is removed and generally it is used to create bevelled edge.

Shaper → Quick return mechanism :-



NOTE:

Working principle of shaper :- It is a process in which your job is stationary and cutting tool moves in forward & backward dirⁿ.

Quick return mech^m :- It is a mech^m on which most of the shaper machines utilise crank and slotted lever to convert rotary motion into reciprocating motion. It consists of rotating crank, slider, slotted lever and connecting rod.

(1) Crank :- It is driven by motor, rotates at a const. speed.

(2) Slider :- It is connected to the crank. It slides within the slotted lever.

(3) Slotted lever :- It oscillates as the slider moves.

(4) Connecting rod :- It connects slotted lever to frame.

(5) CW

Applications of lathe :-

- (i) It is generally used for machining engine components.
- (ii) Also used for wooden products and in wood working industries.
- (iii) For producing precision parts in various metal working industries.
- (iv) Also helps to repair critical components like pumps, valves in oil and gas industries.
- (v) Also helps to produce and repair unique components which are used in aircraft and automobile industries.

Applications of shaper :-

- ↳ Shaper machine helps to produce dice and punches.
- ↳ Also helps to smooth rough surface on metals and other materials.
- ↳ Also helps to machine irregular and complicated profile jobs.

Applications of planer :-

- ↳ Planer is generally used to flat and shape large and heavy metal parts.
- ↳ It also helps to produce parallel flat surfaces with high precision on large sized plates.
- ↳ It also helps to cut various slots and grooves in metal surface.

Applications of slotted

- ↳ They are generally used to cut internal and external grooves in various components with high precision.
- ↳ It is also used to cut keyways in shaft.
- ↳ It also helps to repair and maintain various types of components in machinery.
- ↳ These types of machines are generally used in rolling mills, paper mills and various ship building industries.

Cutting stroke :- The crank rotates, slider moves within the slotted lever and slotted lever oscillates as a result of which the ram and tool move forward. It is known as cutting stroke (or cutting motion).

Return stroke :- As the crank continues to rotate, the slider moves again, because of slotted lever the return stroke is completed faster than cutting stroke. As the non-productive time is reduced, the productivity of the machine is increased.

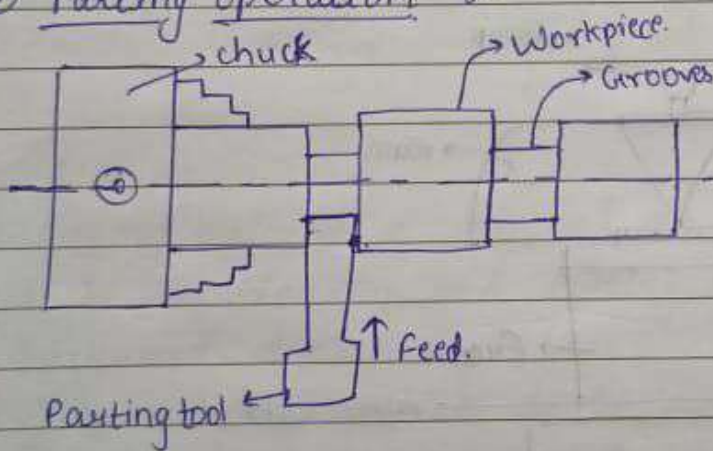
⑤ Boring operation :-

It is defined as enlarging of a pre drill hole is to be formed with the help of boring tool.

NOTE :- $\rightarrow$ For drilling operation boring is not essential but for boring operation, the drilling is to be essential.

$\rightarrow$ As the drill bits of each and every dimension is not available in the market, but sometimes as per customer requirements the different sizes of internal holes is to be formed. To obtain this, the boring operation is to be performed in the metal cutting industries.

⑥ Parting operation :-



Feed $\rightarrow$ mm/revolution

Feed :- It is defined as the distance cutting tool advance into or along the workpiece with each revolution of the workpiece.

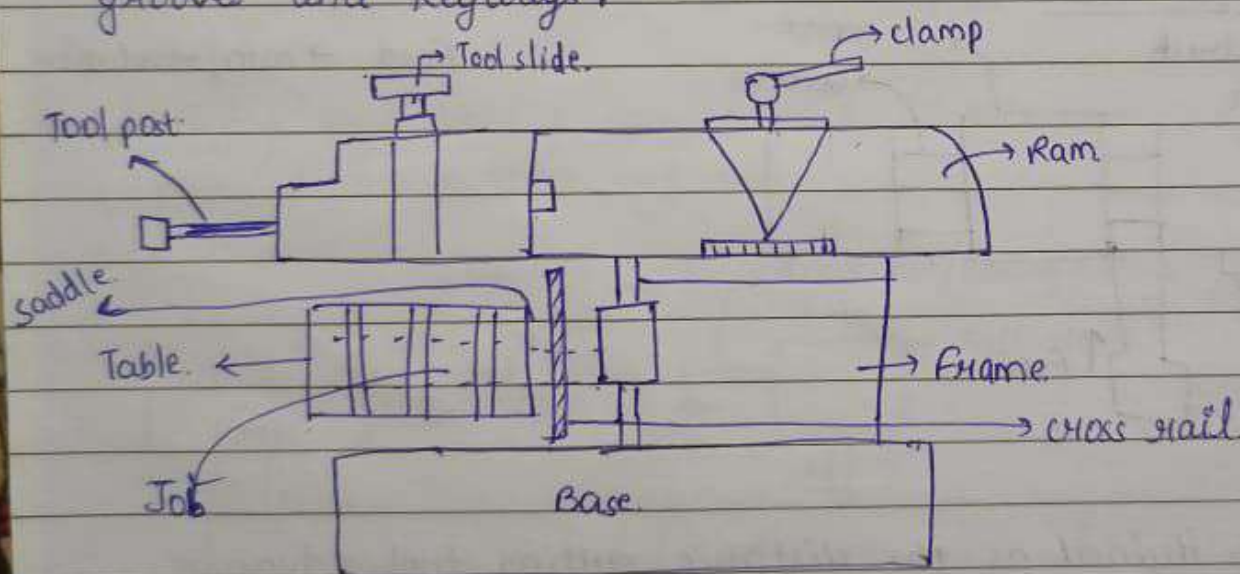
It is an operation in which the standard product is separated off or cutoff from parent material. The tool which is used in this process is known as parting tool.

⑦ Thread cutting operation :-

The helical grooves cut on a cylindrical specimen is called as thread. In this process, the threads are cut in the form of forming operation and the shape of the cutting tool is same as per the thread requirement.

Shaper

It is defined as a machine tool which is very slow and produce low metal removal rate and generally used to produce flat, specific profile on the work piece. Generally it is used to produce slots, grooves and keyways.



→ Shaper works on the principle of quick return.

NOTE: It is generally used for machining flat surface in horizontal, vertical and angular direction. In this machine, the surface is generated by the linear motion of the cutting tool, Job is stationary. Linear motion of the tool is to be performed.

Shaper operations :-

(1) Horizontal cutting :- It is a process in which the horizontal surface of an object is to be machined by shifting the workpiece in the opposite direction of the cutting tool. This process is generally used to produce grooves.

(2) Inclined cutting operation :- In this process, all the cuts are created at an angle other than 90° to the horizontal or vertical plane after moving the workpiece on the table as per desired cutting angle.

(3) Irregular cutting operation :- In this process, the tool utilise with a circular nose and generally this is used to make a curve like profile on the surface of an object.

Applications of shaper machine :-

- They are generally used to create flat surface.
- Also used to smooth the rough surface.
- Used to create keyways, splines and various internal features.
- Also used to produce concave and convex shape on various objects.

NOTE:- It involves reducing the diameter of a rotating workpiece.

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RANKA

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Generally, used to produce shafts, bolts and pins.

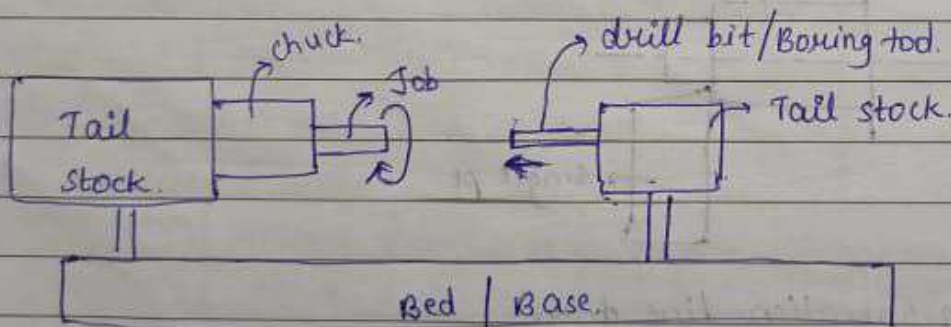
② Facing operation:- (length reduces).

It is a process to generate flat surface in the lathe and the feed is given in a direction $\perp$ ular to the axis of revolution.

③ Knurling operation:-

It is a process in which the grip ~~is~~ is produced on a texture or a pattern is created on the surface of a cylindrical object. It is generally considered in a category of forming operation. It is not considered under cutting operation.

* Drilling and boring operation:- (Inches $\rightarrow$ standardization unit)
8mm $\rightarrow$ drill bit and hole



④ Drilling operation:-

It is an operation in which the cylindrical holes are generated into solid material. The work piece is rotating and the axis of hole is maintained and with the help of drill bits the drilling operation is to be completed.

Applications of slotted :-

- ↳ They are generally used to cut internal and external grooves in various components with high precision.
- ↳ It is also used to cut keyways in shaft.
- ↳ It also helps to repair and maintain various types of components in machinery.
- ↳ These types of machines are generally used in rolling mills, paper mills and various ship building industries.

Difference between centre lathe and capstan and turret lathe :-

Centre lathe.

- ↳ It is manually operated lathe.
- ↳ It has only one tool post, so tool changing time is more.
- ↳ It has tail stock.
- ↳ Only one tool can be fitted in the tail stock.
- ↳ ~~Number~~ Number of speeds is less.
- ↳ Tool changing time is more.
- ↳ The machine should be stopped for changing tool.
- ↳ It is not suitable for mass production.
- ↳ No feed stops to control the tool.

Capstan and turret lathe.

- ↳ It is semi-automatic lathe.
- ↳ Front and rear tool posts are available. Tool changing time is less.
- ↳ It has turret head instead of tail stock.
- ↳ Six different tools can be fitted in the turret head.
- ↳ Number of speeds is more.
- ↳ Tool changing time is less.
- ↳ Tool can be changed without stopping the machine.
- ↳ It is suitable for mass production.
- ↳ The tools are controlled by feed stops.

- ↳ The tool is centered manually after changing the tool.
- ↳ Only one operation is done at a time.

- ↳ The tool is centered automatically.
- ↳ More than one operation can be done at a time.

* Difference between capstan lathe & Turret lathe *

Capstan lathe

- ↳ It is a light duty machine.
- ↳ The turret head is mounted on the ram and the ram is mounted on the saddle.
- ↳ The saddle will not be moved during machining.
- ↳ The lengthwise movement of turret is less.
- ↳ Short work piece only can be machined.
- ↳ It is easy to move the turret head as it slides over the ram.
- ↳ The turret head cannot be moved crosswise.
- ↳ As the construction of the lathe is not rigid, heavy cut cannot be given.
- ↳ It is used for machining work piece upto ~~100mm~~ 60mm diameter.

Turret lathe

- ↳ It is a heavy duty machine.
- ↳ The turret head is directly mounted on the saddle and the saddle slides over the bed ways.
- ↳ The saddle is moved along with the turret head during machining.
- ↳ The lengthwise movement of turret is more.
- ↳ Long work piece can be machined.
- ↳ It is difficult to move the turret head along with the saddle.
- ↳ The turret head can be moved crosswise in some turret lathes.
- ↳ As the construction of lathe is rigid, heavy cut can be given.
- ↳ It is used for machining work piece up to 200 mm diameter.